2023.11秋线性代数期中(回忆版)

1. Multiple Choice. Only one choice is correct.

(1) Suppose that $\alpha_1,\alpha_2,\alpha_3$ are a basis for the nullspace of a matrix A, N(A). Which of the following lists of vectors is also a basis for N(A)?

(A) $\alpha_1+\alpha_2-\alpha_3,\ \alpha_1+\alpha_2+5\alpha_3,\ 4\alpha_1+\alpha_2-2\alpha_3$.

(B) $\alpha_1+2\alpha_2+\alpha_3,\ 2\alpha_1+\alpha_2+2\alpha_3,\ \alpha_3+\alpha_1+\alpha_2$.

(C) $\alpha_1+\alpha_2,\ \alpha_1+\alpha_2+\alpha_3$

(D) $\alpha_1-\alpha_2,\ \alpha_2-\alpha_3,\ \alpha_3-\alpha_1$

(2) Which of the following statements is correct?

(A) If the columns of A are linearly independent, then $Ax=b$ has exactly one solution for every b.

(B) Any 5×7 matrix has linearly dependent columns.

(C) If the columns of a matrix A are linearly dependent, so are the rows.

(D) The column space and row space of a 10×12 matrix may have differet dimensions.

(3) Let $\alpha_1=\begin{bmatrix}1\\4\\1\end{bmatrix}$, $\alpha_2=\begin{bmatrix}2\\1\\-5\end{bmatrix}$, $\alpha_3=\begin{bmatrix}6\\2\\-16\end{bmatrix}$, $\beta=\begin{bmatrix}2\\t\\3\end{bmatrix}$

β can be writtan as a linear combination of $\alpha_1,\alpha_2,\alpha_3$ if $t=$( )

(A) 1 (B) 3 (C) 6 (D) 9

(4) Which of the following statements is correct?

(A) Suppose that $EA=B$ and E is an invertible matrix, then the column space of A and the column space of B are the same.

(B) Let A be n×n matrix with rank 1, then $A^n=cA$, where n is a positive integer and c is a real number.

(C) Let A,B be symmetric matrices, then AB is symmetric.

(D) If A is of full row rank, then $Ax=0$ has only the zero solution.

(5) Let A and B be two n×n matrices. If A is a non-zero matrix and $AB=0$, then

(A) $BA=0$ (B) $B=0$ (C) $(A+B)(A-B)=A^2-B^2$ (D) rank $B<n$.

2. Fill in the blanks.

(1) Denote the vector space of 7×7 real matrices by $M_{7\times7}(\mathbb{R})$, and let W be the subspace of $M_{7\times7}(\mathbb{R})$ consisting of skew-symmetric real matrices, then dim W= _____.

A matrix A is called skew symmetric if $A^T=-A$.

(2) Let A,B be two n×n ~~inter~~ invertible matrices. Suppose the inverse of $\begin{bmatrix}A & C\\0 & B\end{bmatrix}$ is $\begin{bmatrix}A^{-1} & D\\0 & B^{-1}\end{bmatrix}$, where 0 is the n×n zero matrix. Then D= _____.

(3) Let $A=\begin{bmatrix} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{bmatrix}$ with rank$(A)<4$, Then $a=$ _____.

(4) Consider the system of linear equations:

$$Ax=b=\begin{cases} x+2y=1 \\ z-y=2 \\ y=-1 \end{cases}$$

The least-squares solution for the system is _____.

(5) Let H be the subspace of R^3 be defined as follows:

$$H=\left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \middle| x_1+2x_2+x_3=0 \right\}$$

A unit vector orthogonal to H is _____.

3. Consider the following 4×5 matrix A with its reduced row echelon form R:

$$A=\begin{bmatrix} 1 & 2 & * & 1 & * \\ 0 & 1 & * & 1 & * \\ -1 & 1 & * & 3 & * \\ 2 & 0 & * & 1 & * \end{bmatrix}, R=\begin{bmatrix} 1 & 0 & 1 & 0 & 3 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) Find a basis for each of the four fundamental subspaces of A.

(b) Find the third column of matrix A.

4. Let $A=\begin{bmatrix} 1 & -1 & -1 \\ 2 & a & 1 \\ -1 & 1 & a \end{bmatrix}$, $B=\begin{bmatrix} 2 & 2 \\ 1 & a \\ -a-1 & -2 \end{bmatrix}$

For which valus(s) of a, the matrix equation $AX=B$ has no solution, a unique solution, or infinitely many solutions? Where X is a 3×2 matrix. Solve $AX=B$ if it has at least one solution

5. Let $M_{2\times2}(R)$ be the vector space of 2×2 real matrices. Let

$$A=\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, B=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, C=\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

Consider the map

$$T: M_{2\times2}(R)\to R^3, T(X)=\begin{bmatrix} \mathrm{tr}(A^TX) \\ \mathrm{tr}(B^TX) \\ \mathrm{tr}(C^TX) \end{bmatrix}$$

for any 2×2 real matrix X, where $\mathrm{tr}(D)$ denotes the trace of a matrix D.
The trace of an $n\times n$ matrix D is defined to be the sum of all the diagonal entries of D, in other words,

$$\mathrm{tr}(D)=d_{11}+d_{22}+\cdots+d_{nn}.$$

(a) Show that T is a linear transformation.

(b) Find the matrix representation of T with respect to the ordered basis

$$v_1=\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, v_2=\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, v_3=\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, v_4=\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

for $M_{2\times2}(R)$ and the standard basis

$$e_1=\begin{bmatrix}1\\0\\0\end{bmatrix}, e_2=\begin{bmatrix}0\\1\\0\end{bmatrix}, e_3=\begin{bmatrix}0\\0\\1\end{bmatrix}$$

for R^3.

(c) Can we find a matrix X such that $T(X)=\begin{bmatrix}1\\-2\\1\end{bmatrix}$? If yes, please find one such matrix. Otherwise, give an explanation.

6. Let A be an $m\times n$ matrix, B be an $m\times p$ matrix, and C be an $q\times p$ matrix. Show that

$$\operatorname{rank}\begin{bmatrix}A & B\\ 0 & C\end{bmatrix} \geq \operatorname{rank} A + \operatorname{rank} C,$$

where 0 is the $q\times n$ zero matrix.